

Wireshark Packet Analysis

Wireshark: Wireshark is a network protocol analyzer used to capture and inspect packets traveling across a network. It is widely used for:

- Network troubleshooting
- Performance monitoring
- Security analysis
- Protocol analysis
- Debugging communication issues

Protocols to Identify:

- 1) **Ethernet:** Ethernet is the Data Link Layer (Layer 2) protocol responsible for communication within a Local Area Network (LAN).

Information to Observe:

- Source MAC Address
 - Destination MAC Address
 - Ethernet Type
- 2) **ARP (Address Resolution Protocol):** ARP maps an IP address to a MAC address

Information to Record:

- Source IP
- Destination IP
- Protocol
- Packet Size

```

> Frame 2529: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface wlp0s20f3, id 0
> Ethernet II, Src: be:9b:88:1c:9e:3e (be:9b:88:1c:9e:3e), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Address Resolution Protocol (request/gratuitous ARP)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  [Is gratuitous: True]
  Sender MAC address: be:9b:88:1c:9e:3e (be:9b:88:1c:9e:3e)
  Sender IP address: 10.10.32.70
  Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
  Target IP address: 10.10.32.70

```

3) **ICMP (Internet Control Message Protocol):** Used for diagnostics and error reporting

Information:

- Source IP
- Destination IP
- Protocol
- Packet Size

```

> Frame 46: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface wlp0s20f3, id 0
> Ethernet II, Src: Intel_b7:94:ae (60:e3:2b:b7:94:ae), Dst: Fortinet_c8:79:23 (e8:1c:ba:c8:79:23)
> Internet Protocol Version 4, Src: 10.10.34.205, Dst: 142.250.118.113
> Internet Control Message Protocol

```

4) **TCP (Transmission Control Protocol):** Provides reliable communication

Uses the Three-Way Handshake:

SYN

|

SYN ACK

|

ACK

```
› Frame 31: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface wlp0s20f3, id 0
› Ethernet II, Src: Intel_b7:94:ae (60:e3:2b:b7:94:ae), Dst: Fortinet_c8:79:23 (e8:1c:ba:c8:79:23)
› Internet Protocol Version 4, Src: 10.10.34.205, Dst: 23.217.111.184
› Transmission Control Protocol, Src Port: 36566, Dst Port: 443, Seq: 124, Ack: 828, Len: 0
```

5) **UDP (User Datagram Protocol):** Provides fast communication without guaranteed delivery.

Used in:

- Video Streaming
- Gaming
- Voice Calls

```
› Frame 10: 87 bytes on wire (696 bits), 87 bytes captured (696 bits) on interface wlp0s20f3, id 0
› Ethernet II, Src: Intel_b7:94:ae (60:e3:2b:b7:94:ae), Dst: Fortinet_c8:79:23 (e8:1c:ba:c8:79:23)
› Internet Protocol Version 4, Src: 10.10.34.205, Dst: 10.10.19.211
› User Datagram Protocol, Src Port: 46988, Dst Port: 53
› Domain Name System (query)
```

6) **HTTP:** Transfers web pages

Information:

- Source IP
- Destination
- Protocol
- Packet Size

```

> Frame 538: 406 bytes on wire (3248 bits), 406 bytes captured (3248 bits) on interface wlp0s20f3, id 0
> Ethernet II, Src: Intel_b7:94:ae (60:e3:2b:b7:94:ae), Dst: Fortinet_c8:79:23 (e8:1c:ba:c8:79:23)
  > Destination: Fortinet_c8:79:23 (e8:1c:ba:c8:79:23)
  > Source: Intel_b7:94:ae (60:e3:2b:b7:94:ae)
  Type: IPv4 (0x0800)
> Internet Protocol Version 4, Src: 10.10.34.205, Dst: 34.223.124.45
> Transmission Control Protocol, Src Port: 44818, Dst Port: 80, Seq: 1, Ack: 1, Len: 340
  Source Port: 44818
  Destination Port: 80
  [Stream index: 20]
  > [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 340]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 3715503845
  [Next Sequence Number: 341 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 433428537
  1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x018 (PSH, ACK)
  Window: 63
  [Calculated window size: 64512]
  [Window size scaling factor: 1024]
  Checksum: 0xcd5d [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  > [Timestamps]
  > [SEQ/ACK analysis]
  TCP payload (340 bytes)
> Hypertext Transfer Protocol

```

7) **HTTPS:** Secure version of HTTP.

Uses:

- SSL
- TLS Encryption

Information:

- Protocol
- Packet Size

8) **DNS (Domain Name System):** Converts domain names into IP addresses.

Information:

- Source IP
- Destination
- Protocol

- Packet Size

```

  Frame 1: 362 bytes on wire (2896 bits), 362 bytes captured (2896 bits) on interface wlp0s20f3, id 0
  Ethernet II, Src: Fortinet_c8:79:23 (e8:1c:ba:c8:79:23), Dst: Intel_b7:94:ae (60:e3:2b:b7:94:ae)
    Destination: Intel_b7:94:ae (60:e3:2b:b7:94:ae)
    Source: Fortinet_c8:79:23 (e8:1c:ba:c8:79:23)
    Type: IPv4 (0x0800)
  Internet Protocol Version 4, Src: 52.110.14.129, Dst: 10.10.34.205
  Transmission Control Protocol, Src Port: 443, Dst Port: 50630, Seq: 1, Ack: 1, Len: 296
    Source Port: 443
    Destination Port: 50630
    [Stream index: 0]
    [Conversation completeness: Incomplete (12)]
    [TCP Segment Len: 296]
    Sequence Number: 1 (relative sequence number)
    Sequence Number (raw): 2788212170
    [Next Sequence Number: 297 (relative sequence number)]
    Acknowledgment Number: 1 (relative ack number)
    Acknowledgment number (raw): 4270069456
    1000 .... = Header Length: 32 bytes (0)
    Flags: 0x018 (PSH, ACK)
    Window: 1021
    [Calculated window size: 1021]
    [Window size scaling factor: -1 (unknown)]
    Checksum: 0xbfc4 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
    [Timestamps]
    [SEQ/ACK analysis]
    TCP payload (296 bytes)
  Transport Layer Security
    TLSv1.2 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
```